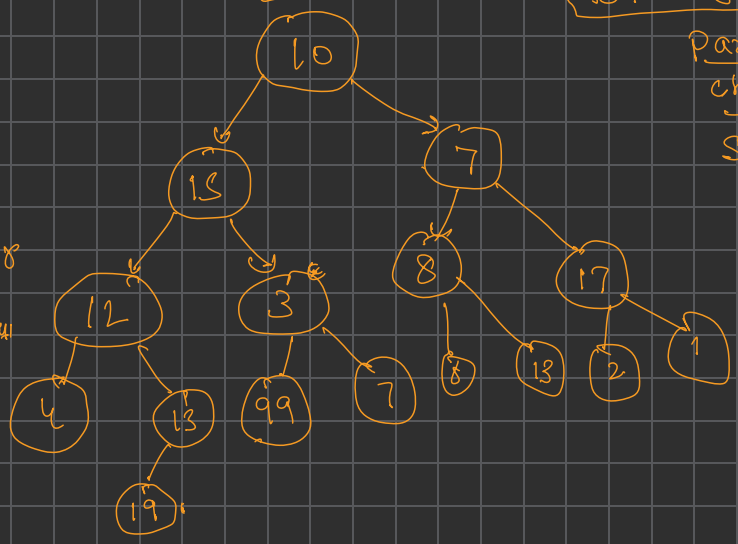
Trees

Binary Tree

Node
root
Leaf
edges
Ancestor
Descendent

Parent
child
Sibling
Level
Height

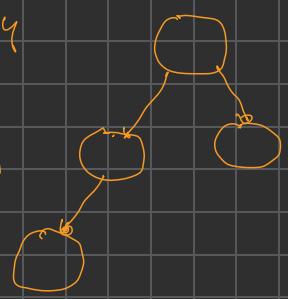


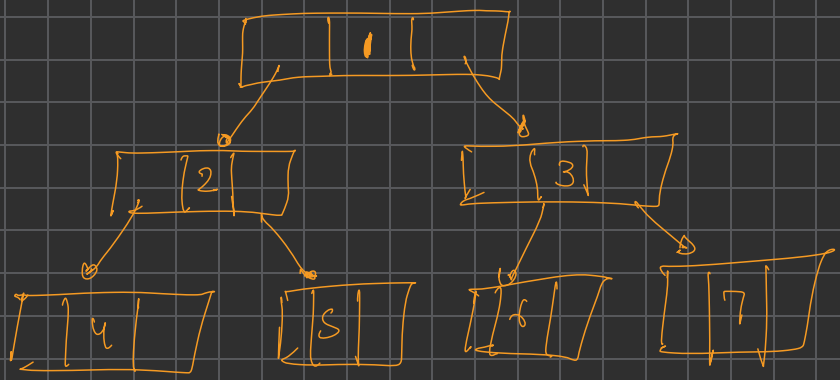
Binary Tree is a tree in which an node has atmost 2 child

Tree: n nodes

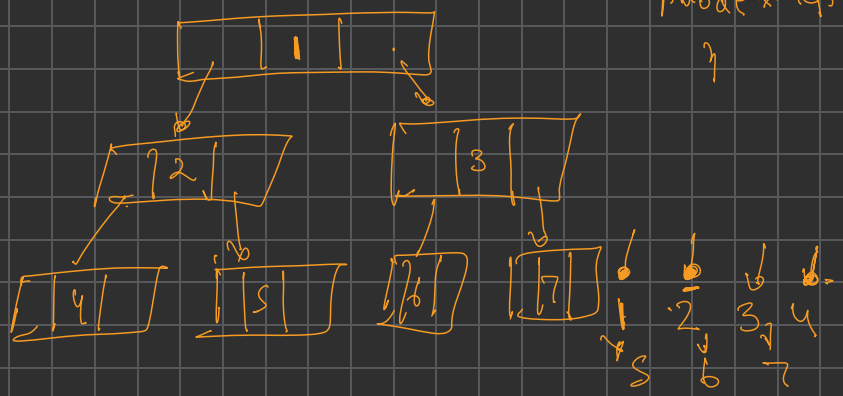
find edges? (n-1) edges

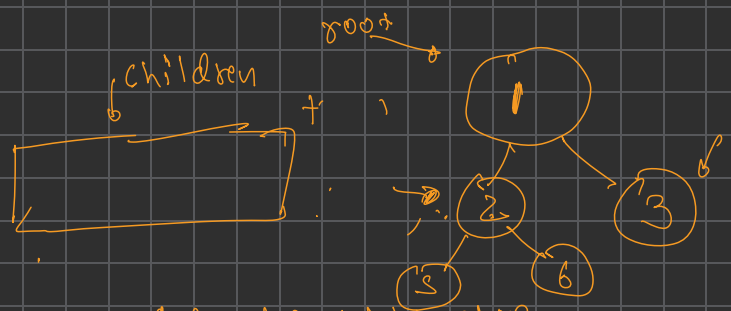
n = ~~2~~ 4
e = ~~1~~ 3

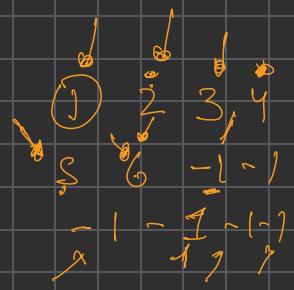





```
Class Node {
  Public
    int data
    Node * left * right
  }
```





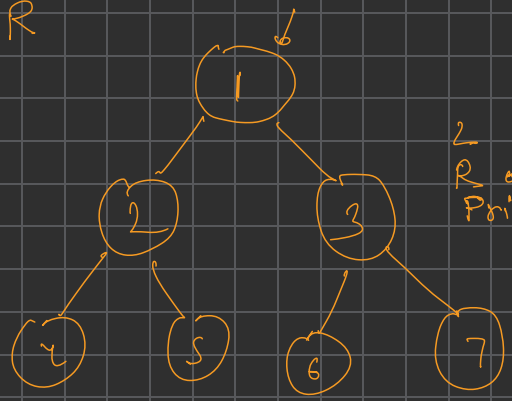



```
int x; leftvalue, rightvalue
cin>>x
Node * root = new Node(x)
queue<Node*> q;
    q.push(root);
while(!q.empty())
{
    Node * temp = q.front();
    q.pop
    cin>>leftvalue;
  if(leftvalue!=-1)
{
temp->left = new Node(leftvalue);
 q.push(temp->left);
}
  cin>>rightvalue;
  if(rightvalue!=-1)
{
  temp->right = new Node(rightval);
  q.push(temp->right);
}
```

6 7

L R

Left
Cout<<root
Right



1. Pre-order
2. Inorder
3. Post-order

L
R
Print

Cout<< node->dat
// Left side
// right side

Preorder : N L R
inorder : L N R
Post or : L R N

```
void preorder(Node * root)
{
    if (root==NULL)
    {
        return;
    }

    Cout<< root->data;
    preorder (root->left);
    Preorder (root->right);

}

void inorder (Node * root)
{
    if (!root)
        return

    inorder (root->left)
    cout<< root->data;
    inorder (root->right);

}
```